

SDS 4135 - Project 3

Due October 18

Grand Blvd Bus Data Follow-up Analysis

Students uncovered some interesting patterns in their exploratory analyses. In this second #70 Grand bus project, you will follow up and do either further visualizations or more formal statistical analyses using models.

For this project, you will have more autonomy in choosing the direction of your analysis, to follow up on a potentially interesting aspect of the transit problems we discussed.

Here are some suggestions for this project:

1. “Zoom in” and visualize bus journeys when things go well, and when delays occur. Make animations of individual bus journeys, and of how the positions of the buses on the route evolve over time, how many passengers they carry, etc.
2. Build models for the travel time of a bus trip, accounting for various factors. Use your models or empirical distributions to produce travel time distributions from a south side location and a north side location to the Grand Transit Center.
3. Investigate how long a passenger needs to wait for the bus to arrive, and how this wait time distribution varies across the route and over the course of the day, week, and year. Also uncover how the schedule varies throughout the day and week.
4. Investigate the relationship between weather and travel time. This will involve getting weather information. Do some “storm case studies” of how the bus trips were affected by major storm events.
5. Find a different external dataset you can pair with the bus data to inform a meaningful analysis. For example census data.
6. Find specific instances of “bus bunching” and visualize them. Make some high quality animations. Determine how often this phenomenon occurs and when it is likely to occur.
7. Look at a particular segment or group of segments in more detail. Can you find stoplight effects? Can you find other idiosyncrasies of the segment that show up in the data?
8. Separate out the reroute period and analyze the differences between the reroute and regular route periods

You are also free to explore some other aspects not listed here.

Please send me what your group wishes to analyze via e-mail before class on Thursday October 8. Also send a backup topic in case we get too many groups choosing the same topic(s).

We will use the preliminary reports to refine your project scope.

Data Usage

Please note and abide by the data usage statement from Metro Transit:

“The accompanying dataset (“70_grand_20250701-20260630.csv”) is provided exclusively for academic instruction. It is not official Metro Transit/Bi-State Development data, nor does it reflect the agency’s system

of record. Students may use this dataset only within the scope of this course. Redistribution or external use is not permitted.”